

[80]: #4. count dulpicates

```
print(df.duplicated().sum())
```

0

[81]: #5. Dropping Duplicate Records

#Before

```
print(df.shape)
```

#Remove duplicates

```
df.drop_duplicates(inplace=True)
```

#After

```
print(df.shape)
```

(52, 6)

(52, 6)

[82]: print(df.columns)

Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')

[83]: #6. String operation

#string methods mark only on text columns

#remove Extra Spaces

```
df['Student_Name']=df['Student_Name'].str.strip()
```

#convert to Uppercase

```
df['Student_Name']=df['Student_Name'].str.upper()
```

```
print(df['Student_Name'].head())
```

#convert to Title case

```
df['Student_Name']=df['Student_Name'].str.title()
```

```
print(df['Student_Name'].head())
```

0 STUDENT1

1 STUDENT2

2 STUDENT3

```

3    STUDENT4
4    STUDENT5
Name: Student_Name, dtype: object
0    Student1
1    Student2
2    Student3
3    Student4
4    Student5
Name: Student_Name, dtype: object

```

```

[85]: df['Department']=df['Department'].str.replace(
        'CSE','Computer Science') #it is only for one val change
print(df['Department'].head())

df['Department']=df['Department'].replace({
    'CSE':'Computer Science',          #for multiple values we use dictionary without string
    'ECE':'Electronics and communi eng',
    'EEE':'Electrical and electronic',
    'IT':'Information Tech',
    'MECH':'Mechanical'
})
print(df['Department'].head()) #for multi values use dict

```

```

0          IT
1          ECE
2          EEE
3          MECH
4    Computer Science
Name: Department, dtype: object
0          Information Tech
1    Electronics and communi eng
2    Electrical and electronic
3          Mechanical
4          Computer Science
Name: Department, dtype: object

```

Name: Department, dtype: object

```
[86]: #String Length
df['Name_Length']=df['Student_Name'].str.len() #name length column is created and the student name length is b
print(df[['Student_Name','Name_Length']].head())
```

	Student_Name	Name_Length
0	Student1	8
1	Student2	8
2	Student3	8
3	Student4	8
4	Student5	8

```
[87]: #Extract First Character
df['First_Letter']=df['Student_Name'].str[0]
print(df[['Student_Name','First_Letter']].head())
```

	Student_Name	First_Letter
0	Student1	S
1	Student2	S
2	Student3	S
3	Student4	S
4	Student5	S

DATA VISUALIZATION

[1]: !pip install matplotlib

```
Requirement already satisfied: matplotlib in c:\users\admin\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (1.2.0)
Requirement already satisfied: cyclor>=0.10 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: numpy>=1.23 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (1.26.4)
Requirement already satisfied: packaging>=20.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\admin\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
```

[5]: `import matplotlib.pyplot as plt #interface`
`x=[1,2,3,4,5]`
`y=[10,20,30,40,50]`



```
plt.plot(x,y) # x axis and y axis import

plt.show() #that is been showed
```

<Figure size 640x480 with 1 Axes> •••

[15]: `#Figure and Axes`
`#These are the most important concepts`
`#Figure-Entire chart window`
`#Axes-The area where the graph is drawn`
`import matplotlib.pyplot as plt`

+

JupyterLab Python [conda env:base]

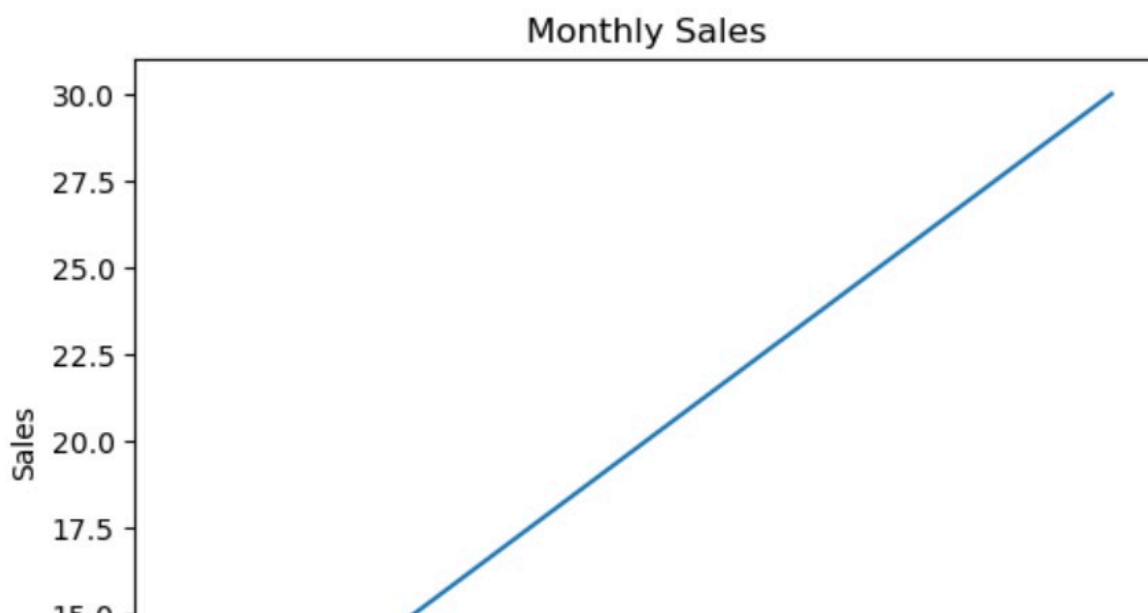
```
fig,ax=plt.subplots()

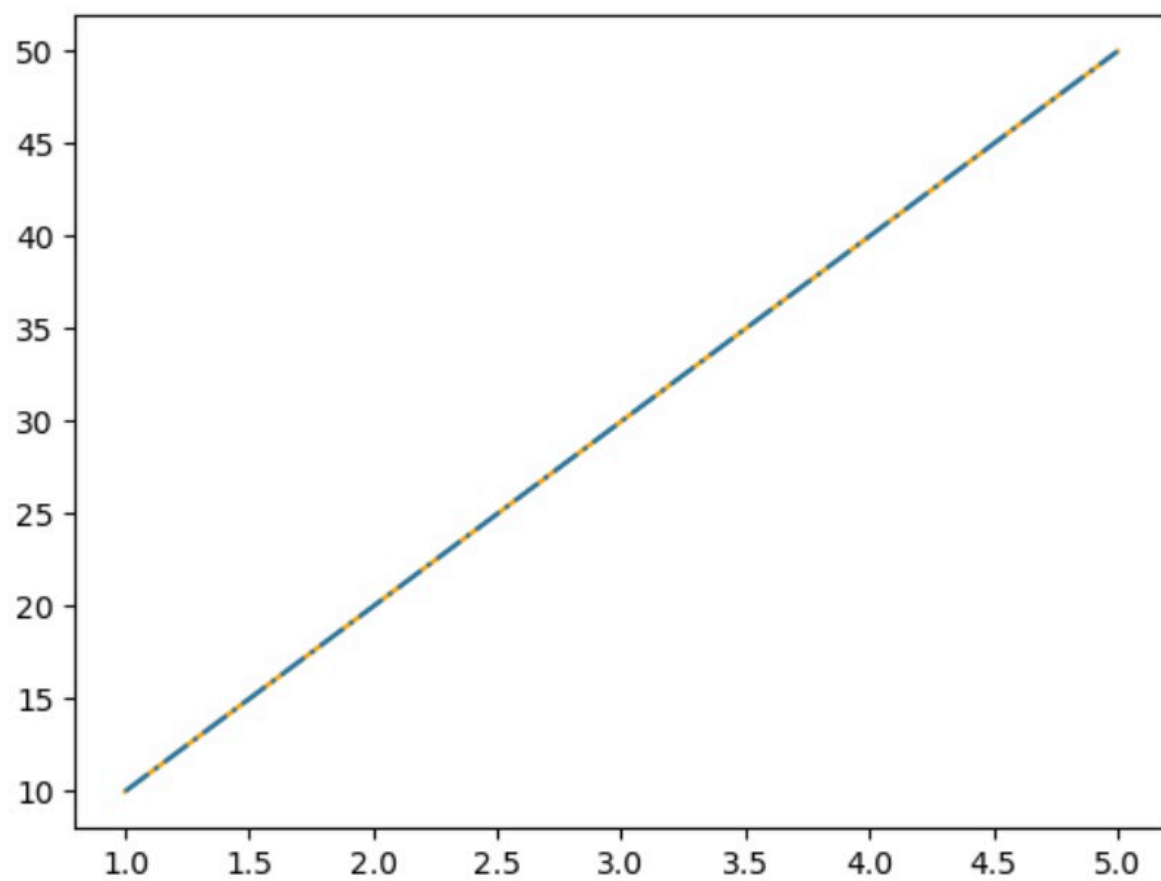
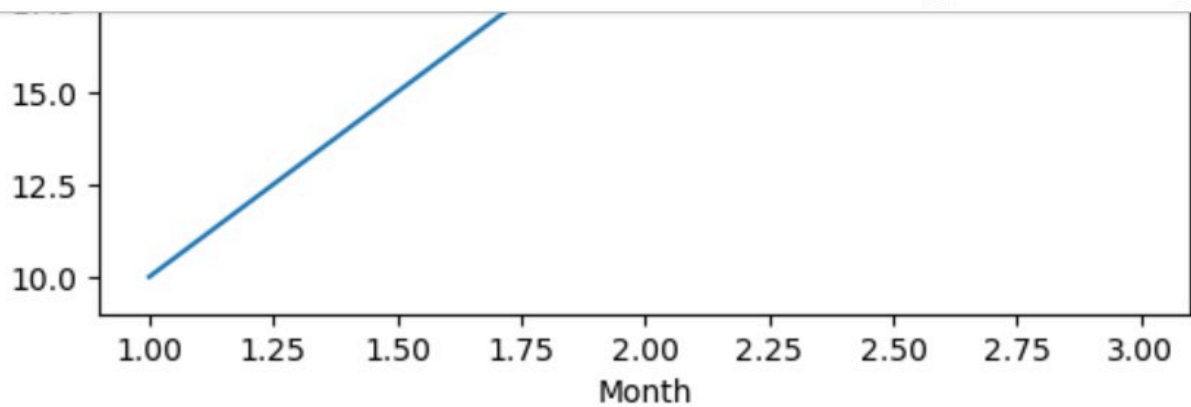
ax.plot([1,2,3], [10,20,30])

#Adding the title
plt.title("Monthly Sales")

#adding x and y labels
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()

#changing line color
plt.plot(x,y,color='orange')
#changing Line Style
plt.plot(x,y,linestyle='-.')
plt.show()
```





```

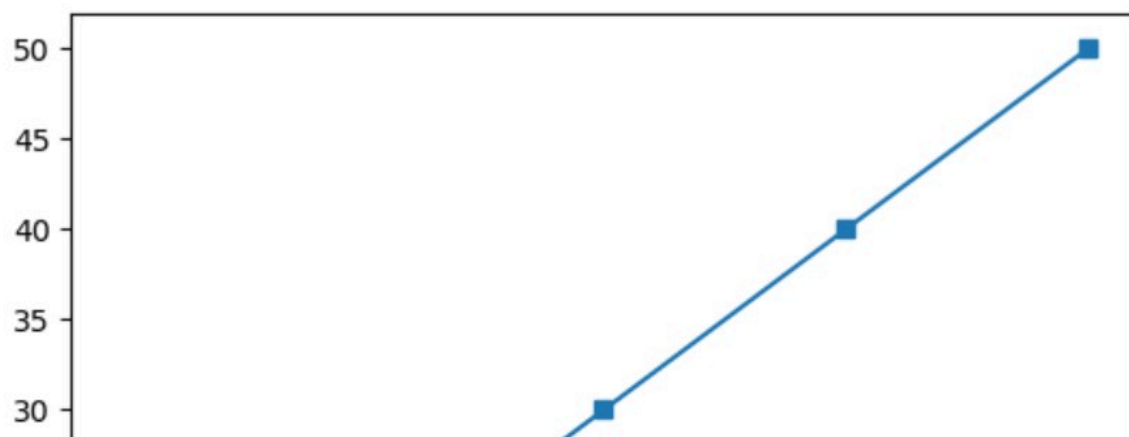
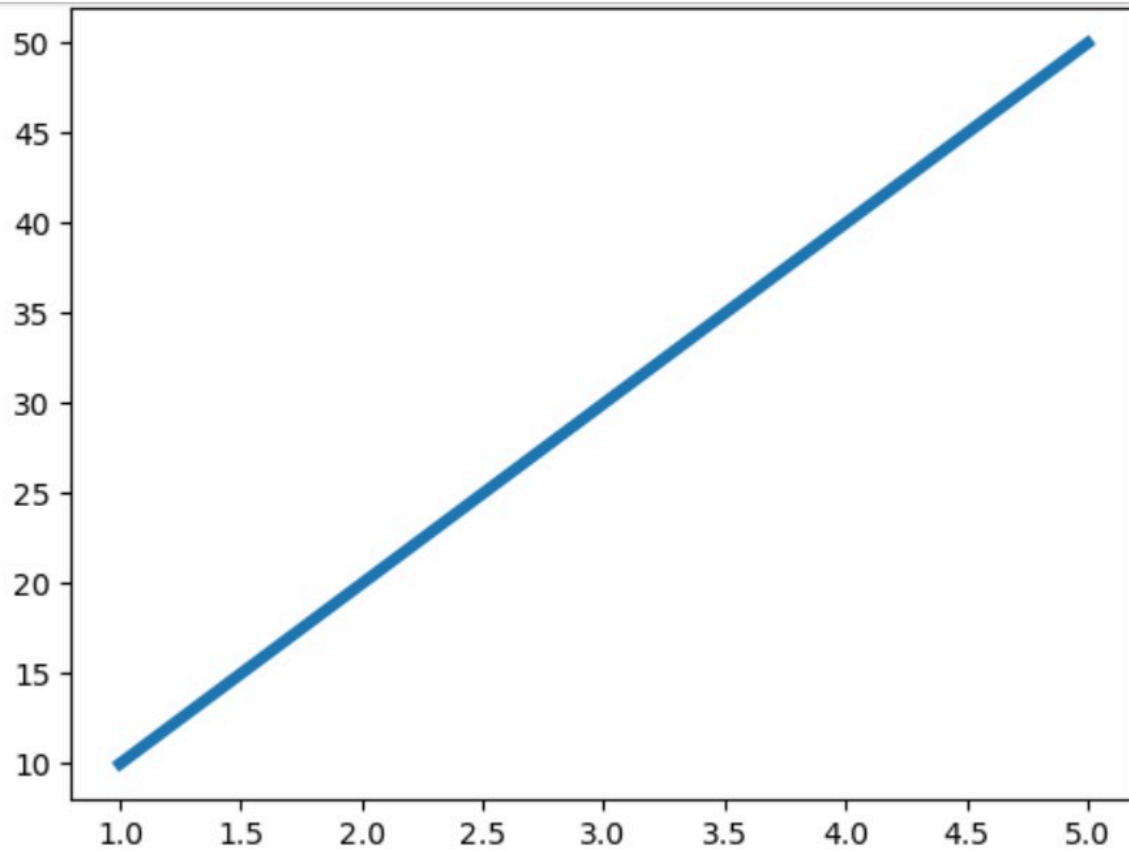
[21]: #Styles:
      #'- ' Solid
      #'-- ' Dashed
      #': ' Dotted
      #'-.' Dash Dot

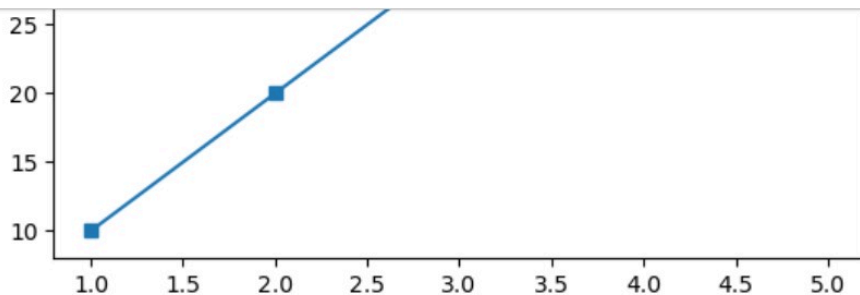
      #Changing Line Width
      plt.plot(x,y,linewidth=4) #line broad
      plt.show()
      #thicker line

      #adding markers
      plt.plot(x,y,marker='s')
      plt.show()

      #output:
      # -----

      #Marker types:
      #'o' Circle
      #'s' Square
      #'^' Triangle
      #'*' Star
  
```



[22]: !pip install seaborn

```
Requirement already satisfied: seaborn in c:\users\admin\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\admin\anaconda3\lib\site-packages (from seaborn) (1.26.4)
Requirement already satisfied: pandas>=1.2 in c:\users\admin\anaconda3\lib\site-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\admin\anaconda3\lib\site-packages (from seaborn) (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\admin\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)
```

```

[33]: import seaborn as sns
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]

#Themes in Seaborn
#Themes contrl the apperance of charts
#Default theme

sns.set_theme()
#Dark Grid theme
sns.set_style("darkgrid")
sns.lineplot(x=x,y=y)
plt.show()

#white background
sns.set_style("white")
sns.lineplot(x=x,y=y)
plt.show()

#ticks background that is same white background
sns.set_style("ticks")
sns.lineplot(x=x,y=y)
plt.show()

#Available styles
#darkgrid
#whitegrid
#dark
#white
#ticks

```

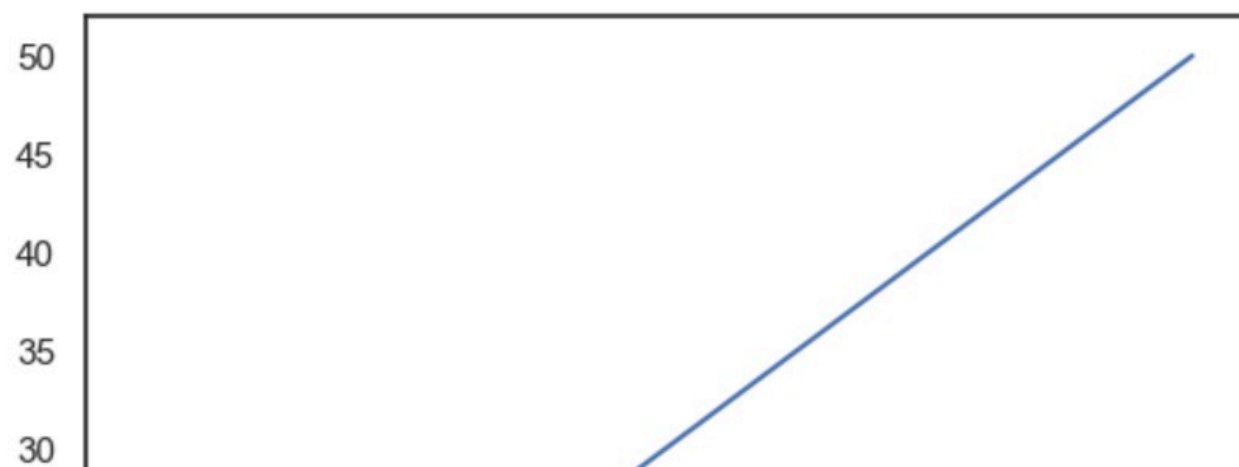
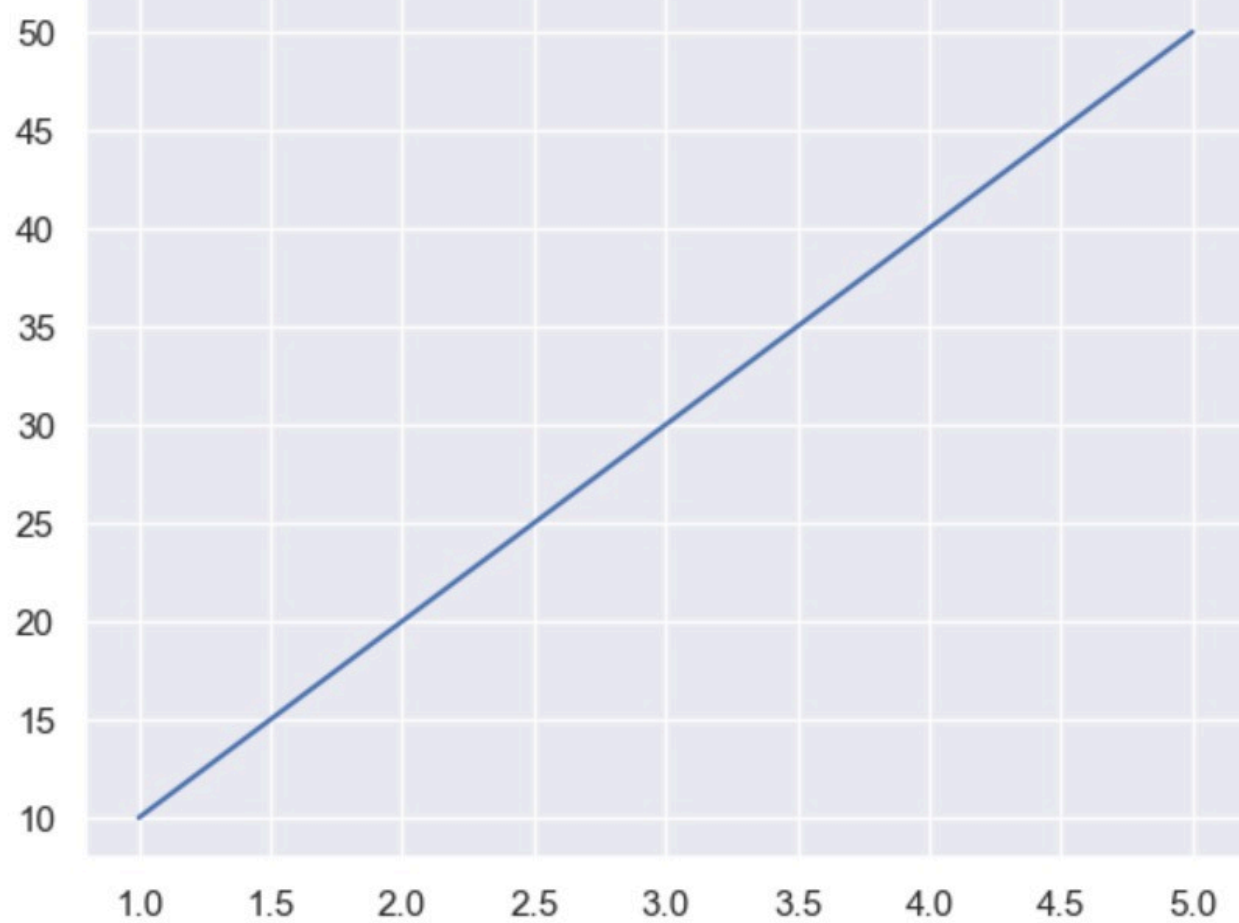
File Edit View Run Kernel Settings Help

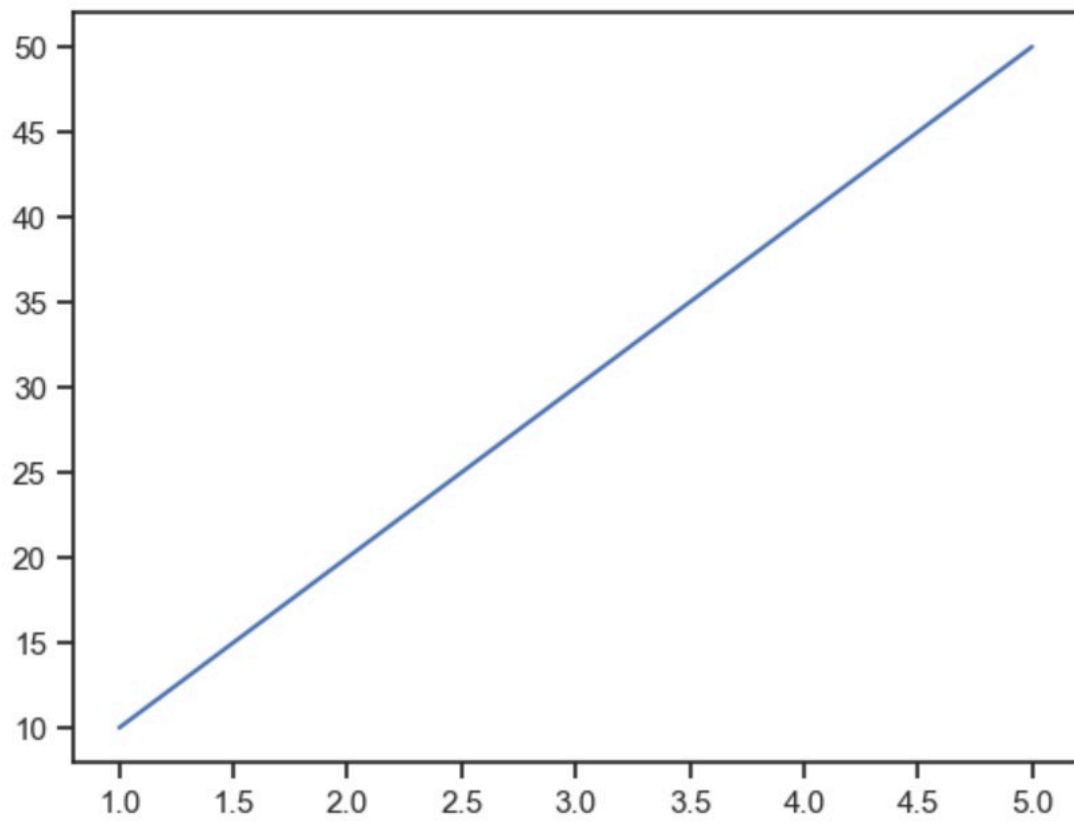
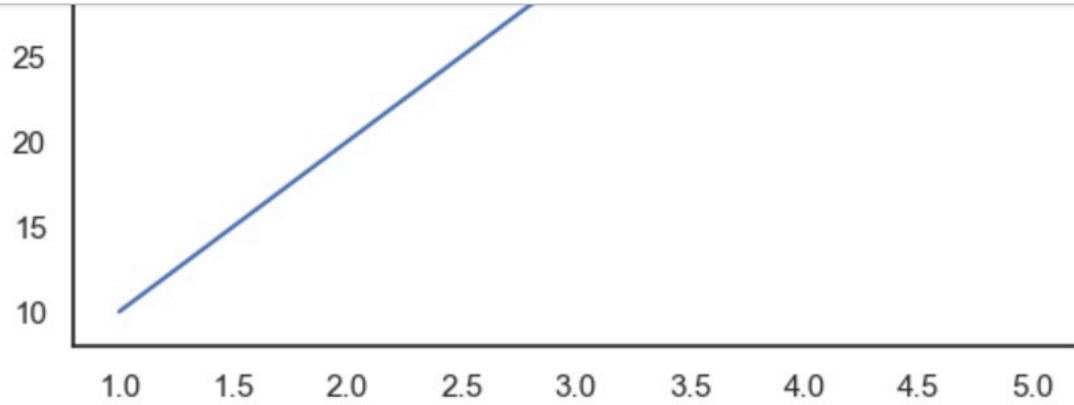
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JupyterLab



Python [c





```
[78]: #Color palatte
      #A palette is a collection of colors

      #View Current Palette
      sns.set_palette("dark") #to change the line of the color

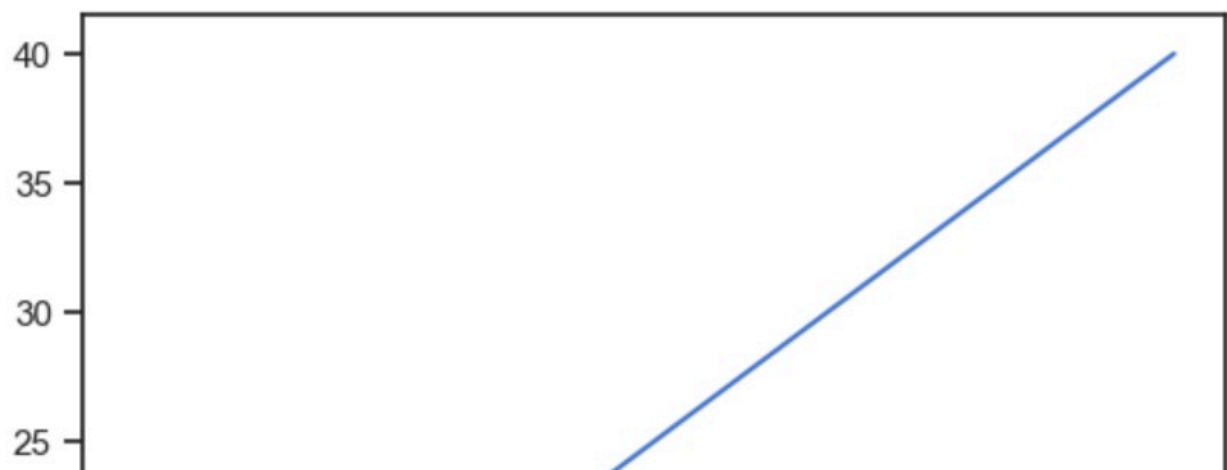
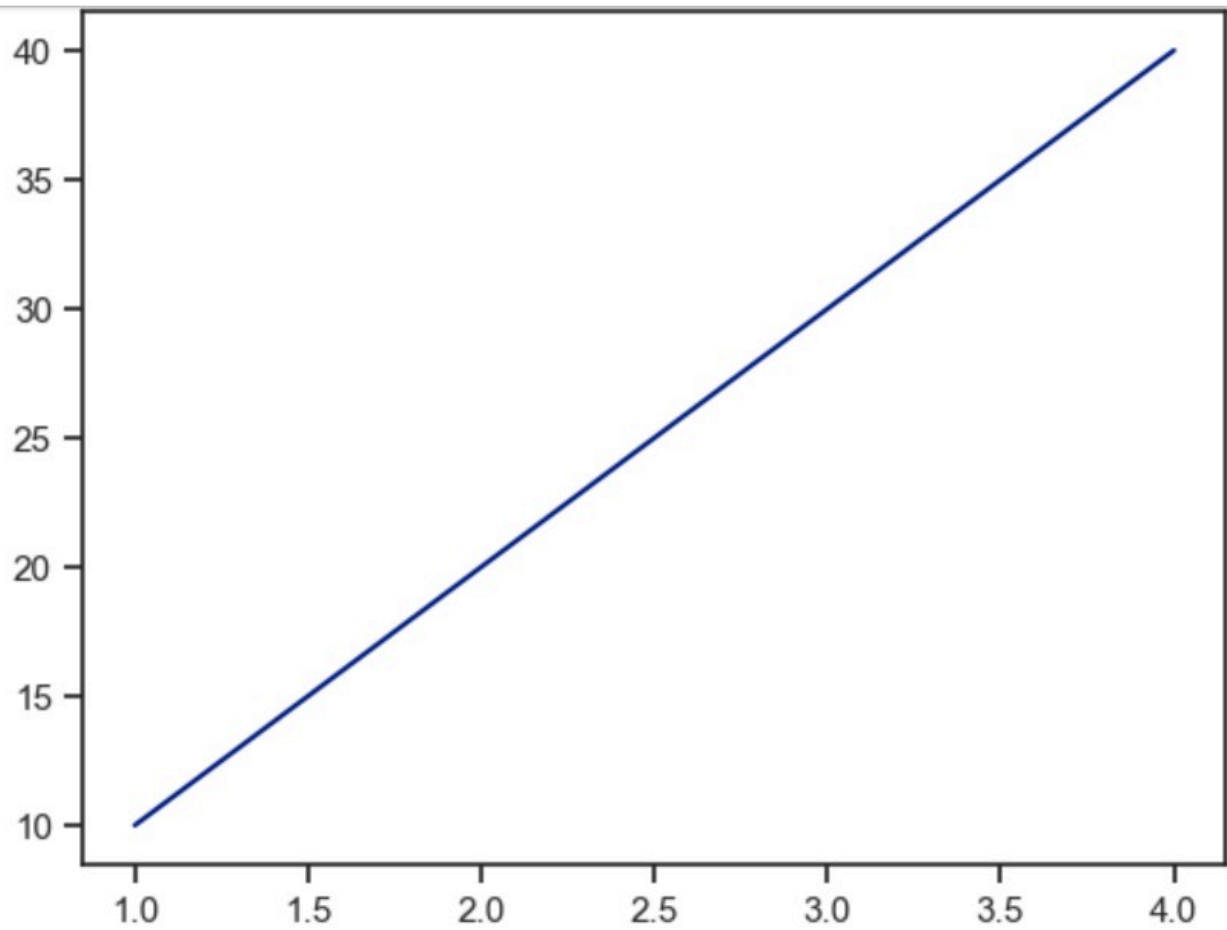
      sns.lineplot(
          x=[1,2,3,4],
          y=[10,20,30,40]
      )
      plt.show()

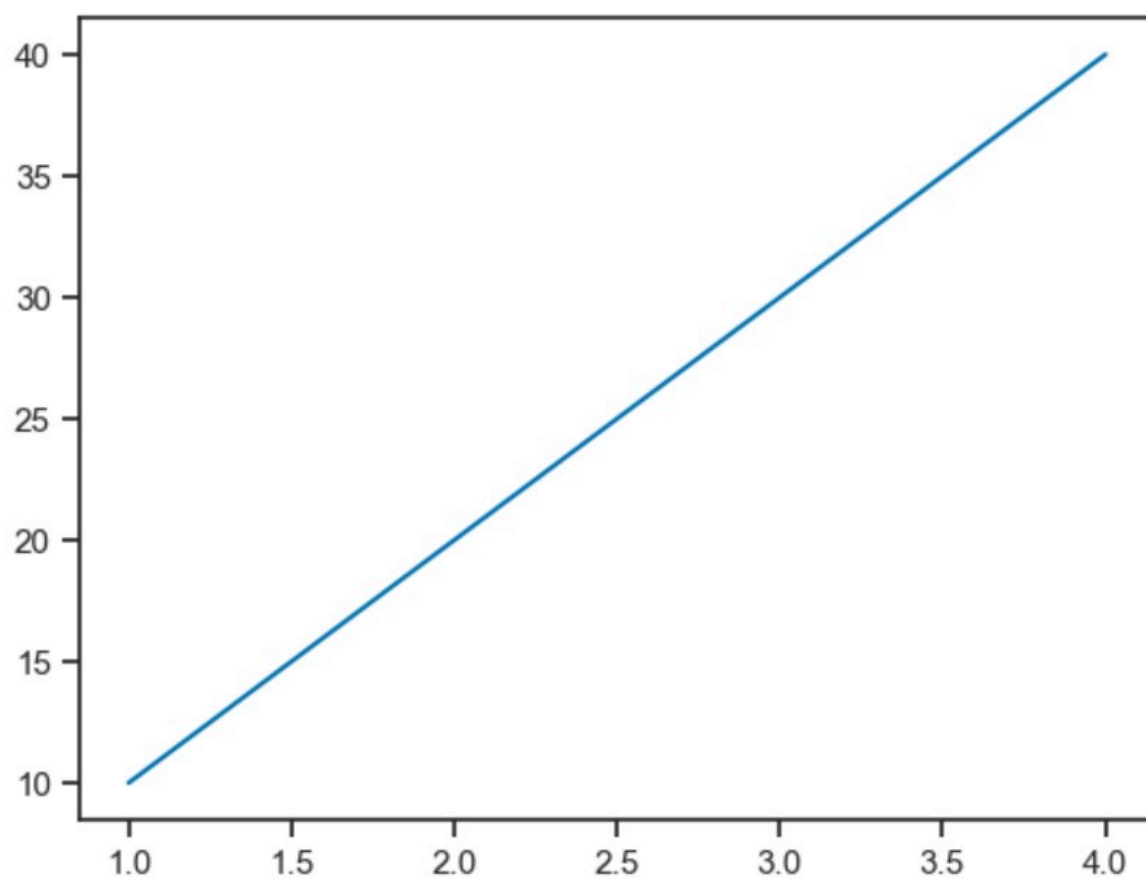
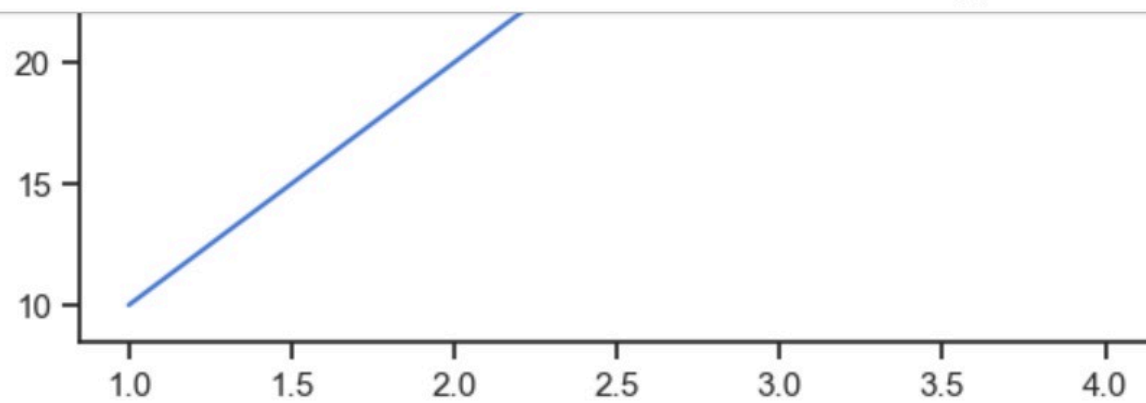
      sns.set_palette("muted") #to change the line of the color

      sns.lineplot(
          x=[1,2,3,4],
          y=[10,20,30,40]
      )
      plt.show()

      sns.set_palette("colorblind") #to change the line of the color

      sns.lineplot(
          x=[1,2,3,4],
          y=[10,20,30,40]
      )
      plt.show()
      #Color available style
      #deep
      #muted
      #bright
      #pastel
      #dark
      #colorblind
```






```
[79]: #Build-in Dataset Support
#One of the Seaborns biggest advantages is build-in datasets
#View Available Datasets
import seaborn as sns
print(sns.get_dataset_names())

['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'flight
s', 'fmri', 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'seaice', 'taxis', 'tips', 'titanic']
```

```
[80]: #Load Titanic dataset
import seaborn as sns
df=sns.load_dataset("titanic")
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
[81]: #Line chart
#What is a Line Chart?
#A line charts are connects data points with lines

#When to use?
#Showing trends over time
```

```
import matplotlib.pyplot as plt
months=['Jan', 'Feb', 'Mar', 'Apr']
sales=[100,150,200,250]

plt.plot(months,sales,marker='o')

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



```
[82]: #Bar chart
#what is a Bar chart?
#a bar chart uses rectangular barrs to compare values

#When to use:
#Comparing categories

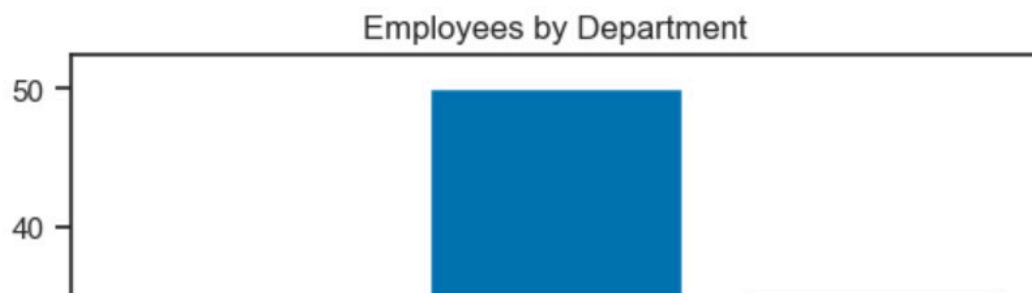
#Examples:
#Department-wise employees
#Product sales
#Student marks
#Population by cities

import matplotlib.pyplot as plt
dept=['HR','IT','Sales']
employees=[20,50,35]

plt.bar(dept,employees)

plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Count")

plt.show()
```



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JupyterLab

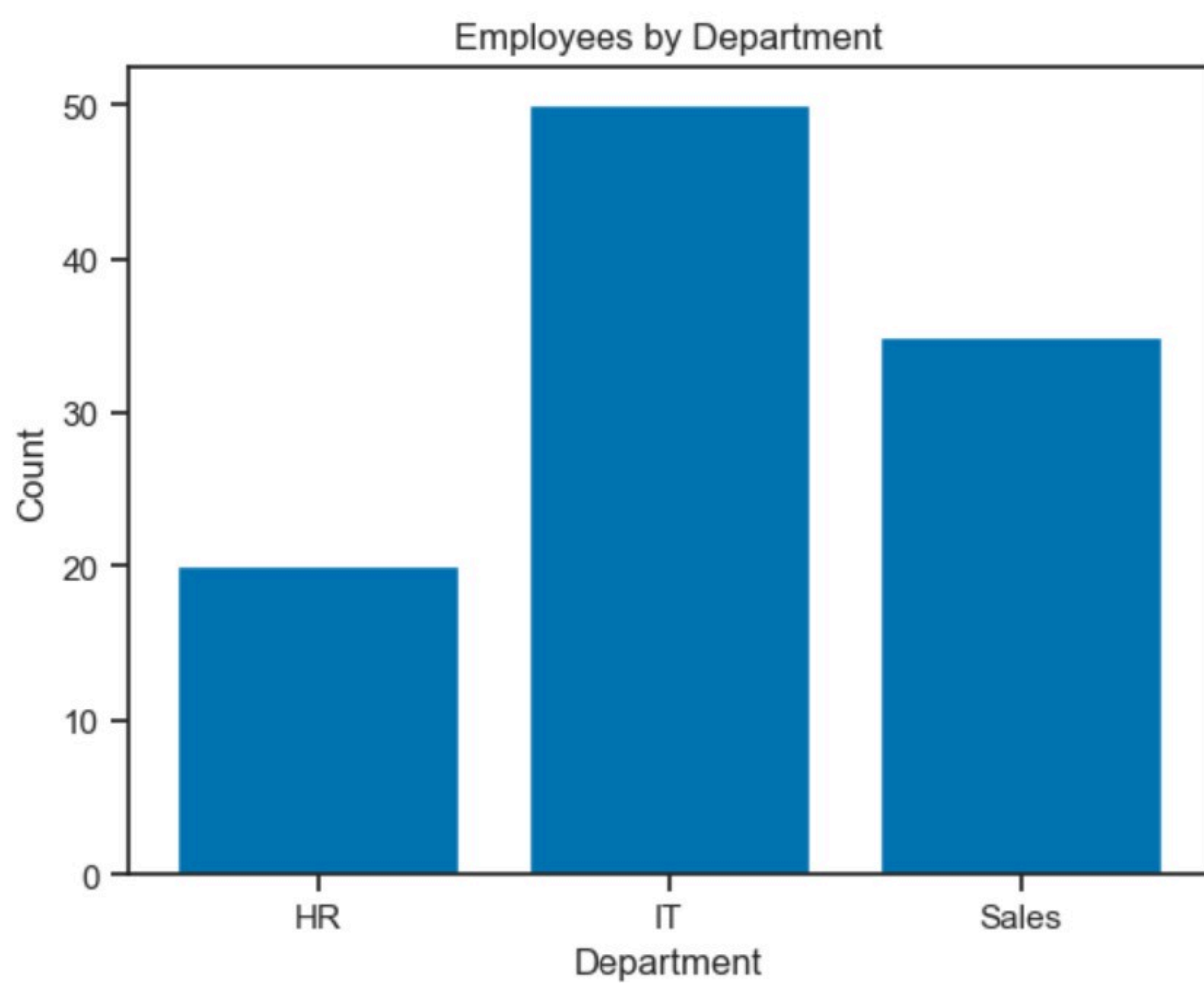


Python [conda env]

```
plt.xlabel("Department")
```

```
plt.ylabel("Count")
```

```
plt.show()
```



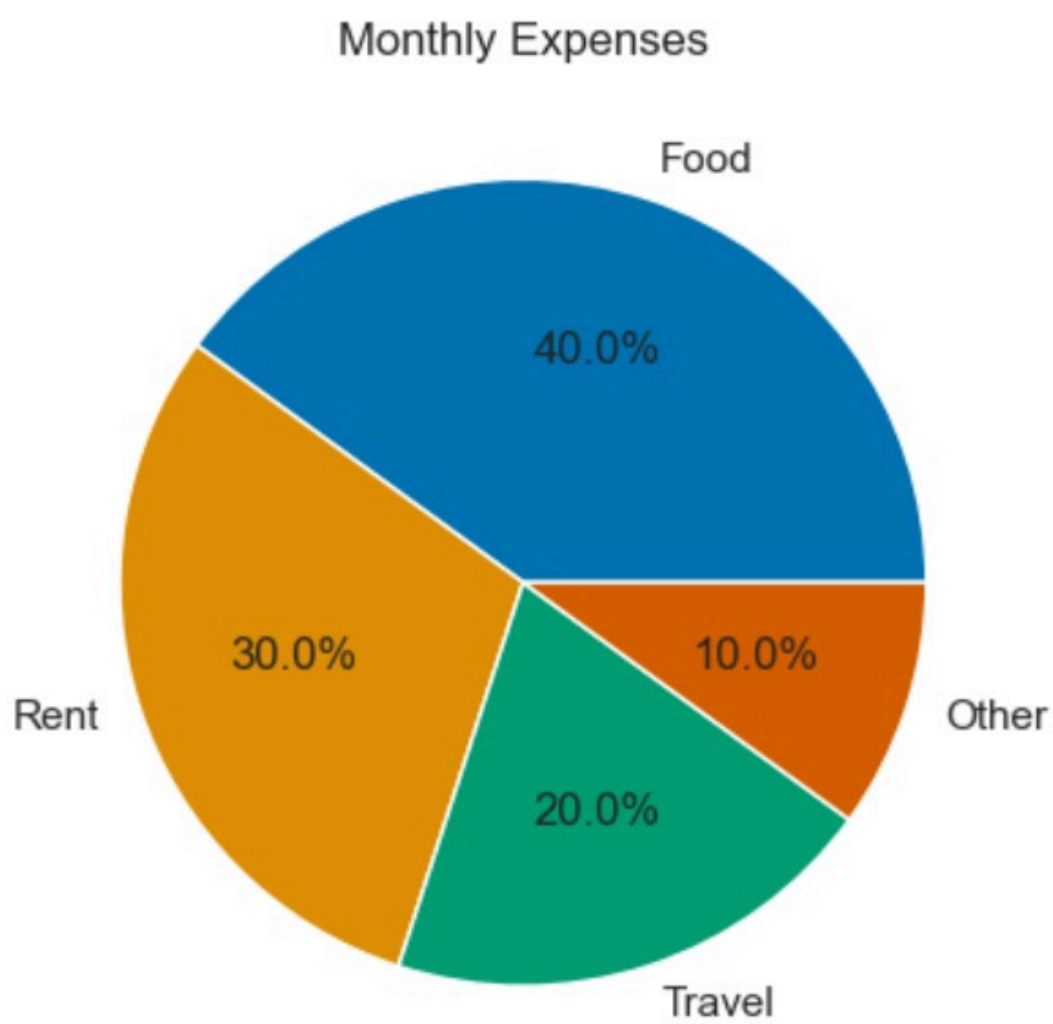
```
[83]: #Pie Chart
      #What is a Pie chart?
      #Pie charts shows the part of a whole

      #when to use:
      #Showing percentages

      #Examples:
      #Budget allocation
      #Market Share
      #Survey results
      #Expense distribution

      import matplotlib.pyplot as plt
      labels=['Food','Rent','Travel','Other']
      values=[40,30,20,10]
      plt.pie(
          values,
          labels=labels,
          autopct='%1.1f%%'    #to show percentage inside the box like 10% inside the part shape
      )
      plt.title("Monthly Expenses")
      plt.show()
```

```
plt.show()
```



```

[84]: #Scatter plot
      #What is Scattter plot?
      #A scatter plot uses dot to represent data points

      #when to use:
      #Finding relationship between two variables

      #Examples:
      #Hight vs weight
      #Study hours vs marks
      #Experience vs Salary

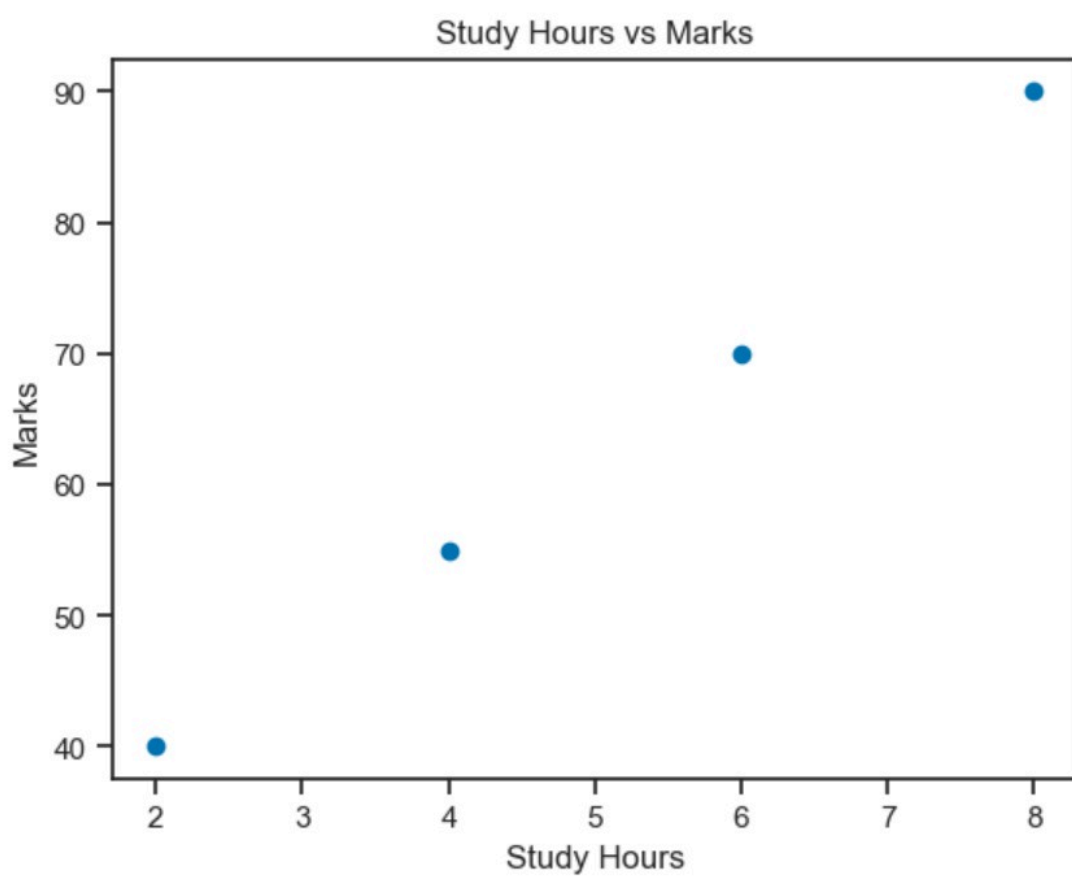
      import matplotlib.pyplot as plt
      hours=[2,4,6,8]
      marks=[40,55,70,90]

      plt.scatter(hours,marks)

      plt.title("Study Hours vs Marks")
      plt.xlabel("Study Hours")
      plt.ylabel("Marks")

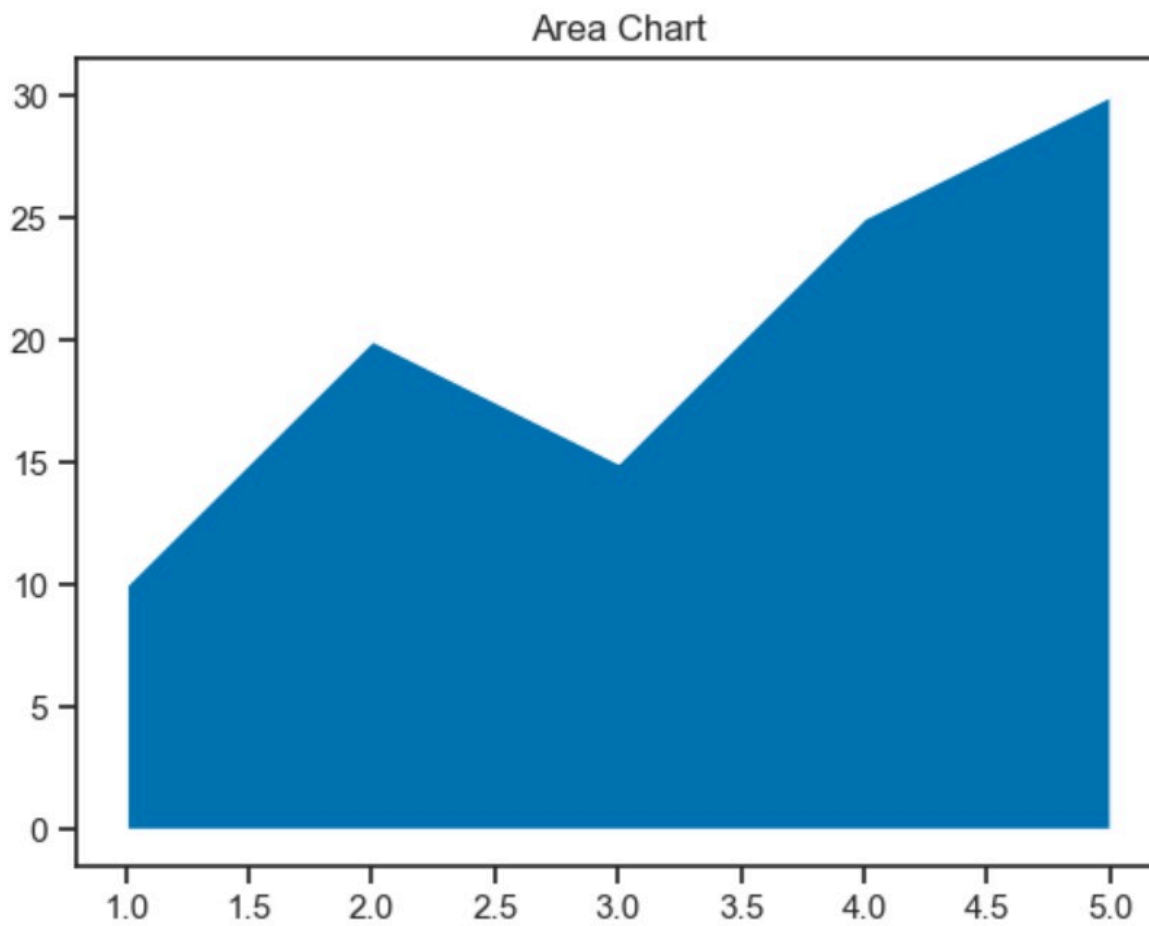
      plt.show()

```

```
[85]: #Area Chart  
      #Purpose: Shows Cumulative totals over time.
```

```
import matplotlib.pyplot as plt  
x=[1,2,3,4,5]  
y=[10,20,15,25,30]  
plt.fill_between(x,y)  
plt.title("Area Chart")  
plt.show()
```



1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0

JupyterLab Python [conda env:ba

```
[86]: #Box plot
#A Box plot is used to visualize the distribution of numerical data and identify outliers

#Components of a Box plot
#min Q1 median Q3 Max
# | |
#Lower Whisker Upper Whisker

#Minimum:Smallest value(excluding outliers)
#Q1(25%):First quartile
#Median(50%):Middle value
#Q3(75%):Third quartile
#Maximum:Largest value(excluding outliers)
#Outliers:Values far from the rest of the data,shown as individual points.

import matplotlib.pyplot as plt
salary=[25000,28000,30000,32000,35000,
        37000,40000,42000,45000,100000]
plt.boxplot(salary)
plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
```

```
plt.show()
```



```
[87]: #Heat Map
#What is heatmap?
# A heatmap uses colors to represent values in a table or matrix
# In data analysis , heatmaps are commonly used to visualize corelations between numerical columns

#Correlation values
#Value meaning
#+1 Strong positive Relationship
#0 No Relationship
#-1 Strong Negative Relationship

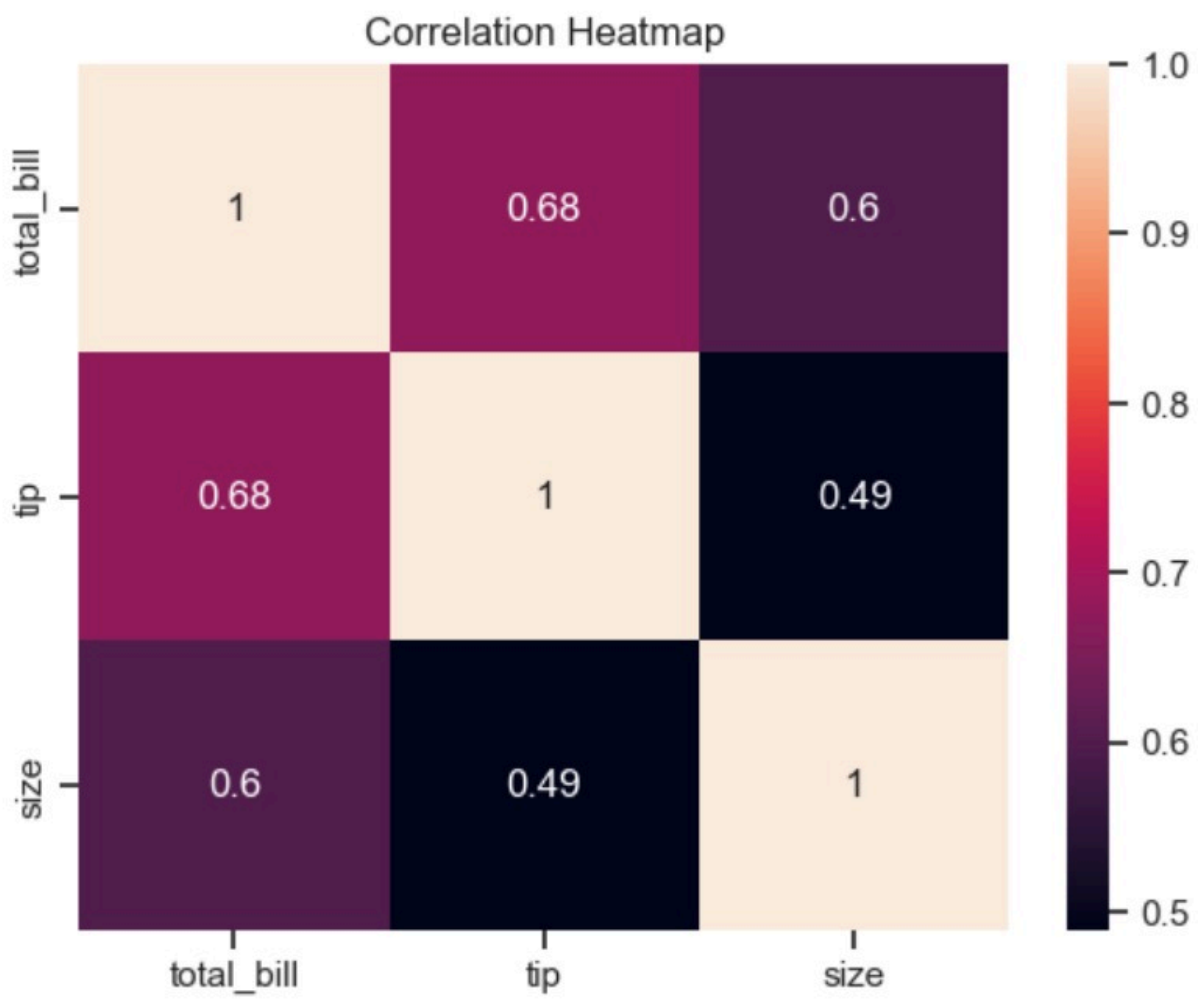
import seaborn as sns
import matplotlib.pyplot as plt

#Load dataset
tips=sns.load_dataset("tips")

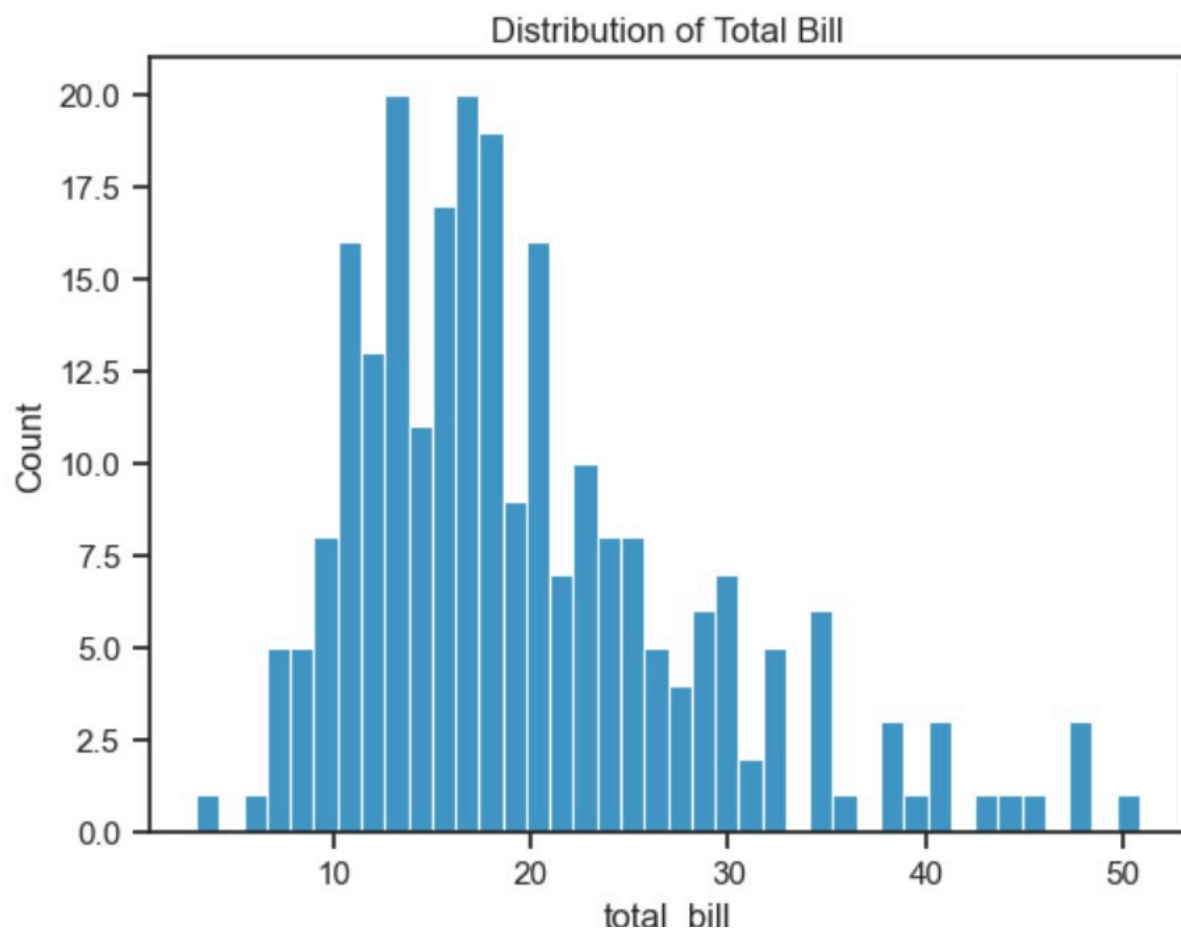
#correlation matrix
corr=tips.corr(numeric_only=True)

#Heatmap
sns.heatmap(corr,annot=True)

plt.title("Correlation Heatmap")
plt.show()
```



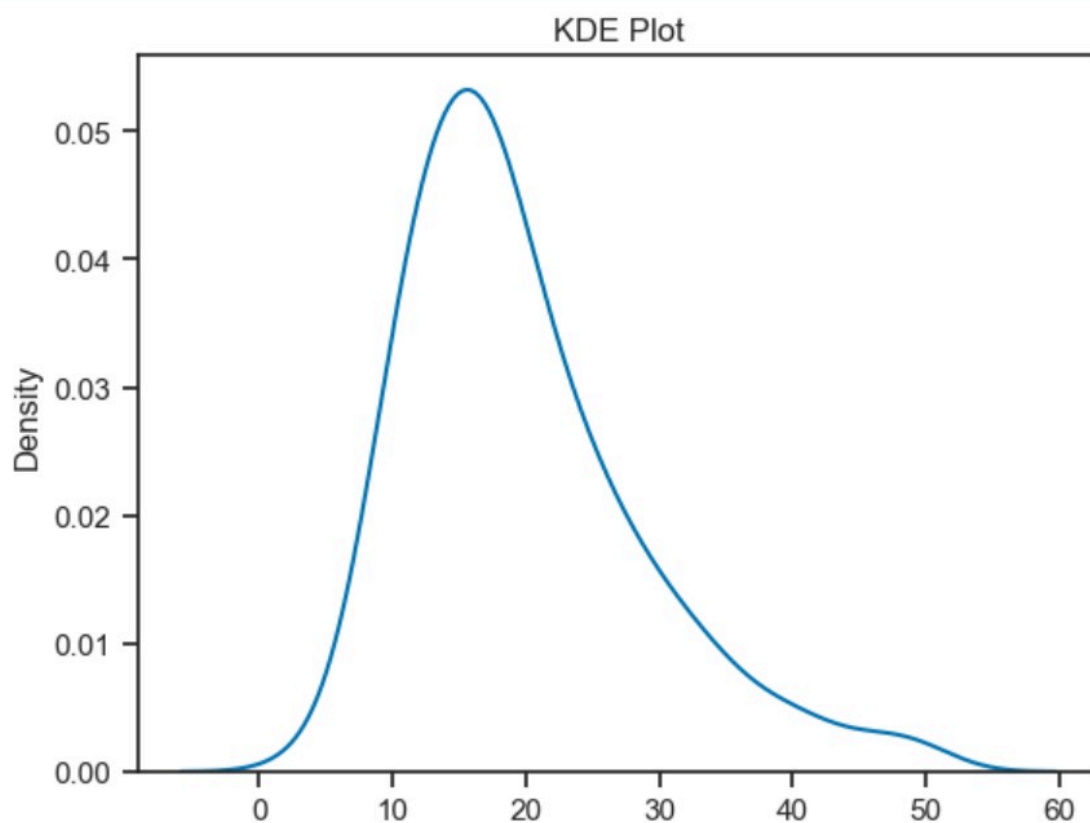
```
#Histogram using Seaborn
sns.histplot(
    data=tips,
    x="total_bill",
    bins=40
)
plt.title("Distribution of Total Bill")
plt.show()
```



```

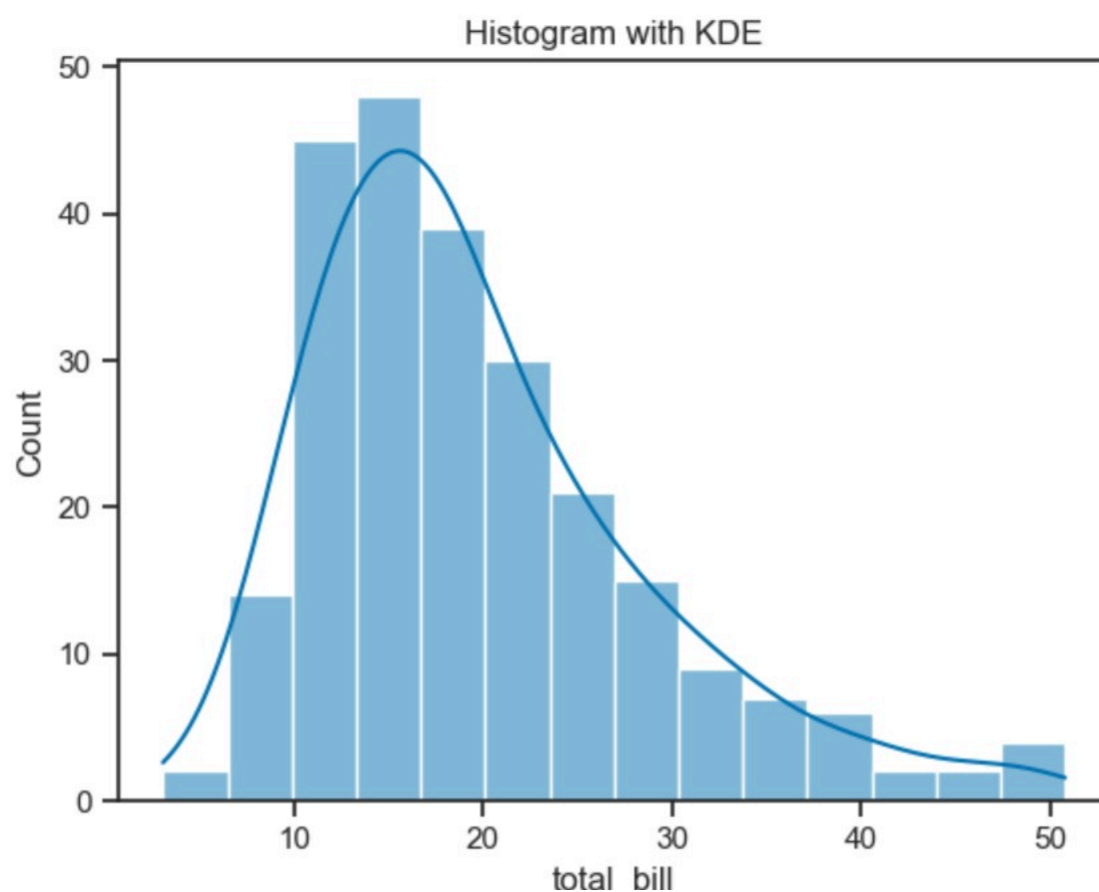
#What is KDE?
#Insted of bars, it shows a smooth curve representing data distribution
#When to use:
#Understanding data distribution identifying peaks comparing distributions Statistical analysis
sns.kdeplot(
    data=tips,
    x="total_bill",
)
plt.title("KDE Plot")
plt.show()

```




```

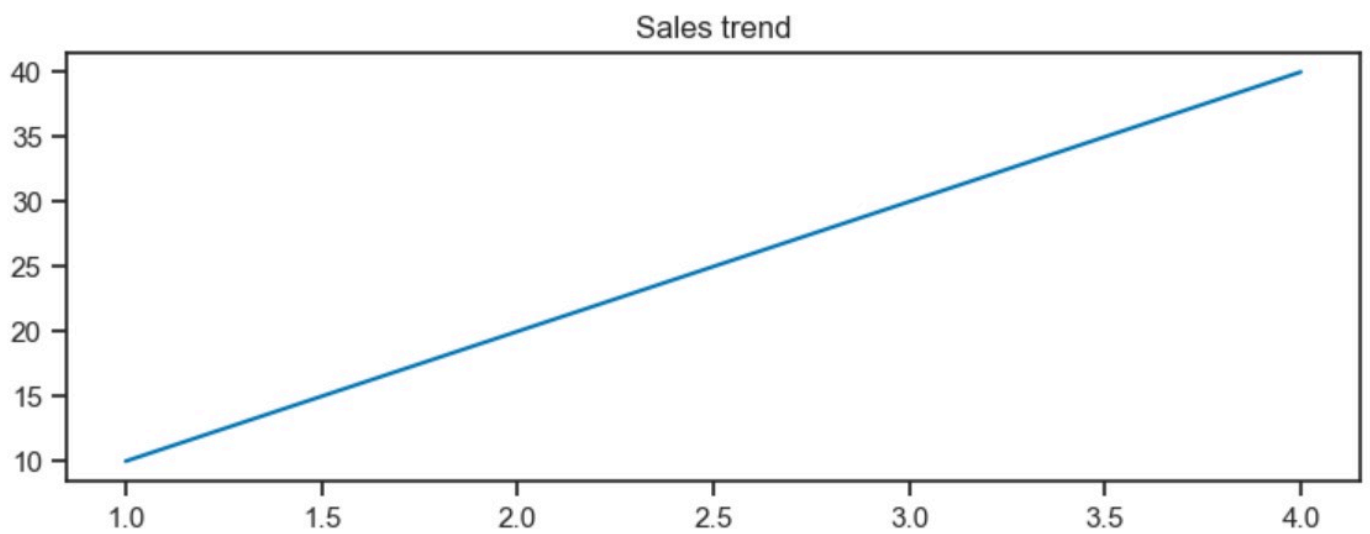
•[101]: #Histogram + KDE Together
        #Most commonly used in Data Science
        sns.histplot(
            data=tips,
            x="total_bill",
            kde=True
        )
        plt.title("Histogram with KDE")
        plt.show()
  
```



10 20 30 40 50
total_bill

```
[111]: #Figure Sizeing
#What is Figure Size?
#Figure Size controls the width and hight of a chart

import matplotlib.pyplot as plt
plt.figure(figsize=(9,3))
plt.plot([1,2,3,4],[10,20,30,40])
plt.title("Sales trend")
plt.show()
```



```

[112]: #Subplots
        #What is subplot?
        #Subplots allow multiple charts in a single figure.
        #Syntax: plt.subplot(rows,columns,position)

        import matplotlib.pyplot as plt

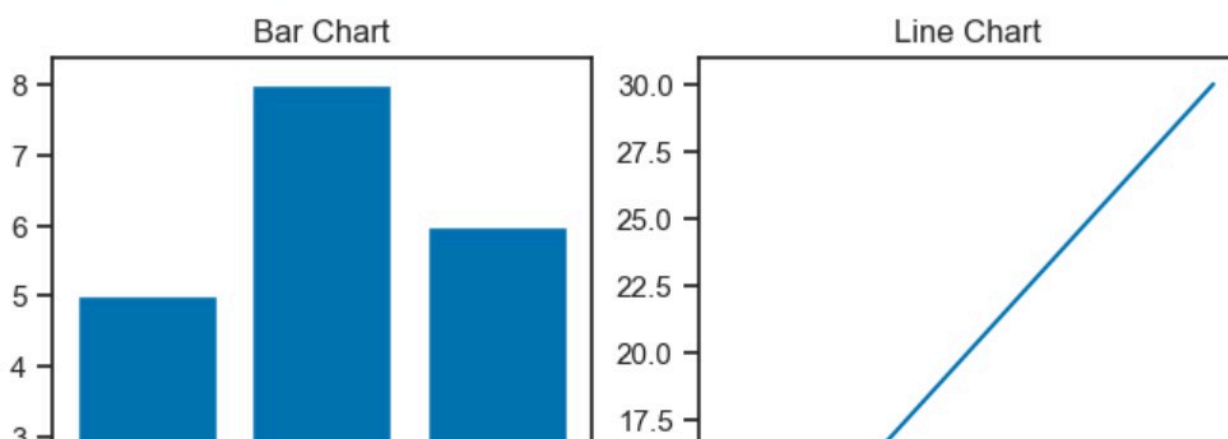
        plt.figure(figsize=(8,4))

        # First Plot
        plt.subplot(1,2,2)
        plt.plot([1,2,3], [10,20,30])
        plt.title("Line Chart")

        # Second Plot
        plt.subplot(1,2,1)
        (plt.bar)(['A','B','C'], [5,8,6])
        plt.title("Bar Chart")
        (plt.show)
    
```

```

[112]: <function matplotlib.pyplot.show(close=None, block=None)>
    
```



```

[118]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))

plt.subplot(2,2,1)
plt.plot([1,2,3], [10,20,30])

plt.title("Line")

plt.subplot(2,2,2)

(plt.bar)(['A','B','C'], [5,8,6])

plt.title("Bar")

plt.subplot(2,2,3)

plt.scatter([1,2,3], [4,8,6])

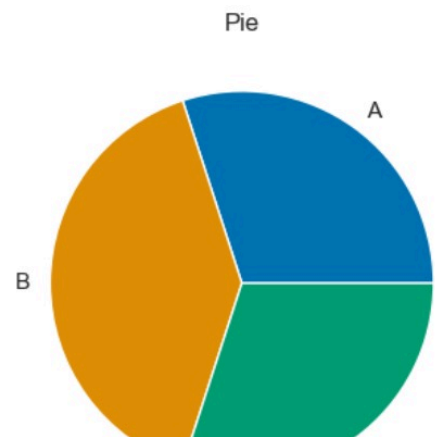
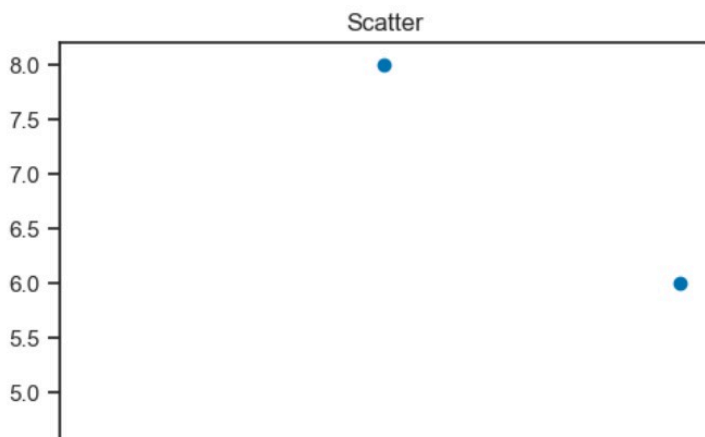
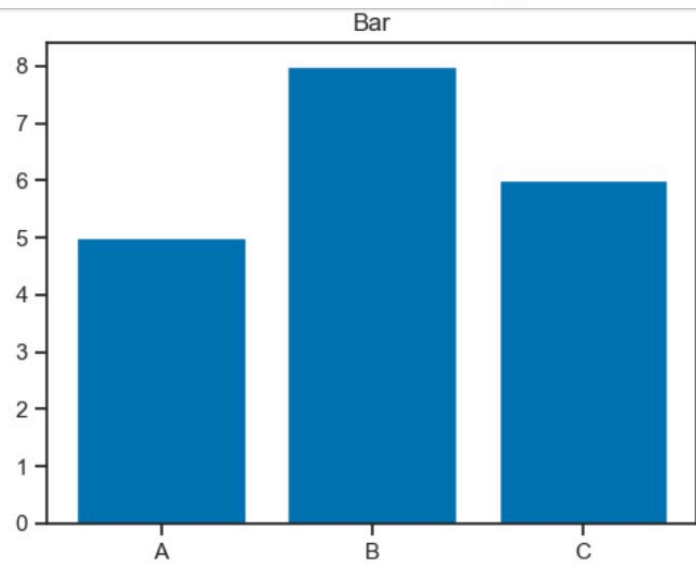
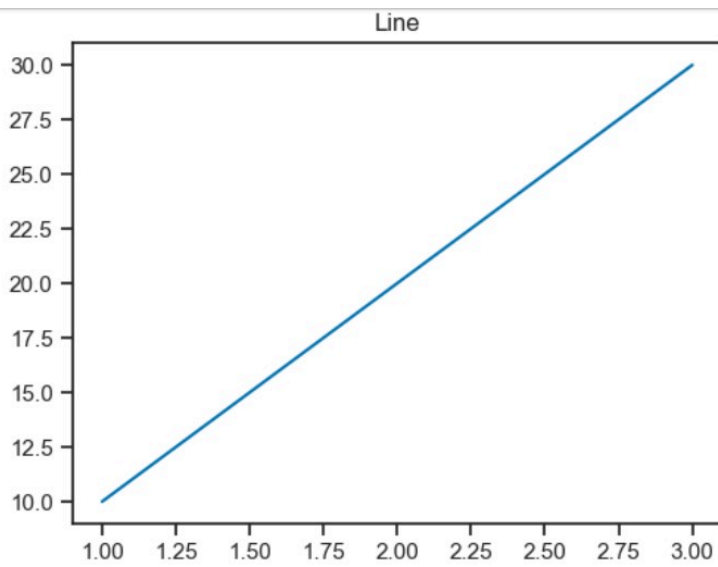
plt.title("Scatter")

plt.subplot(2,2,4)

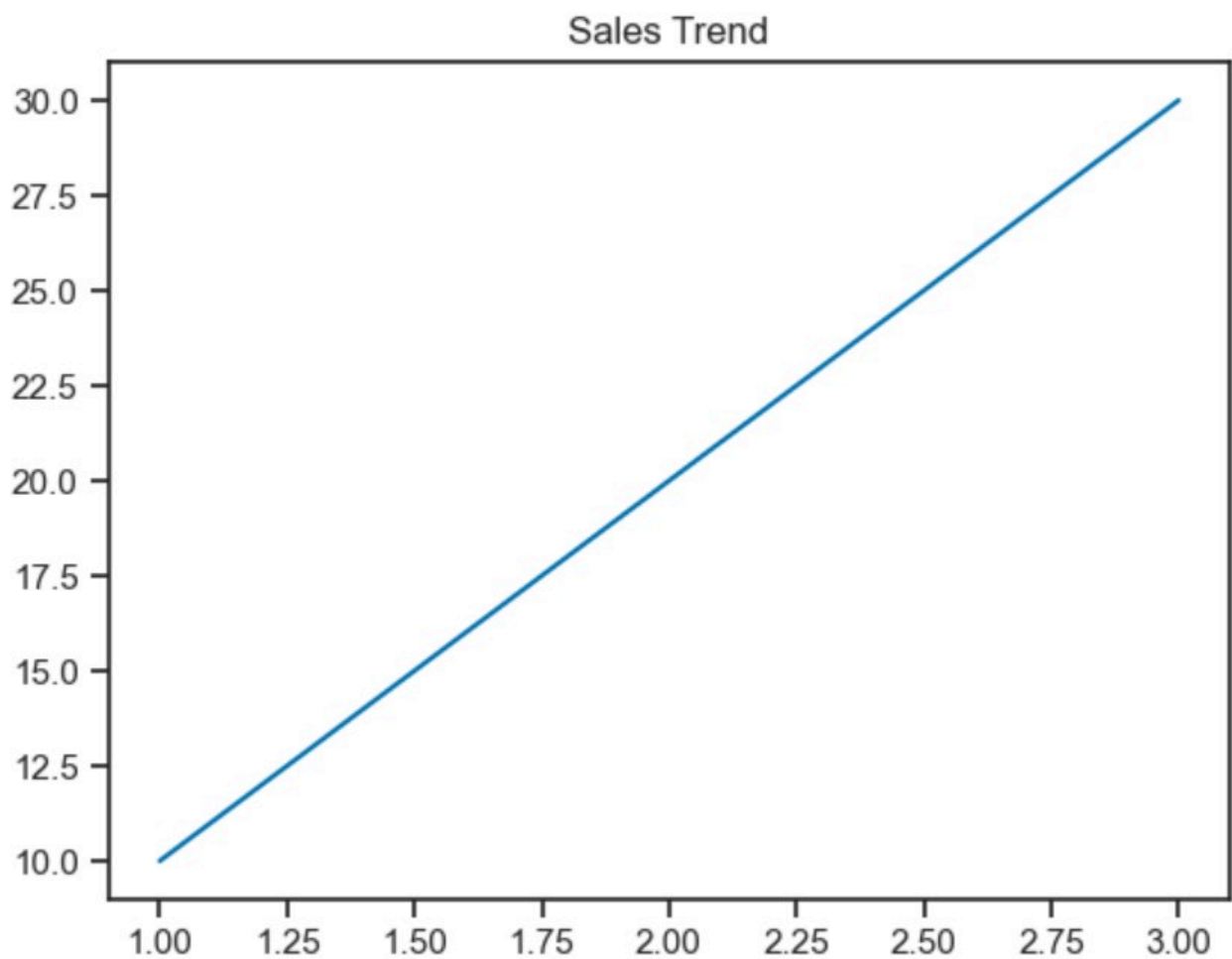
plt.pie([30,40,30], labels=['A','B','C'])

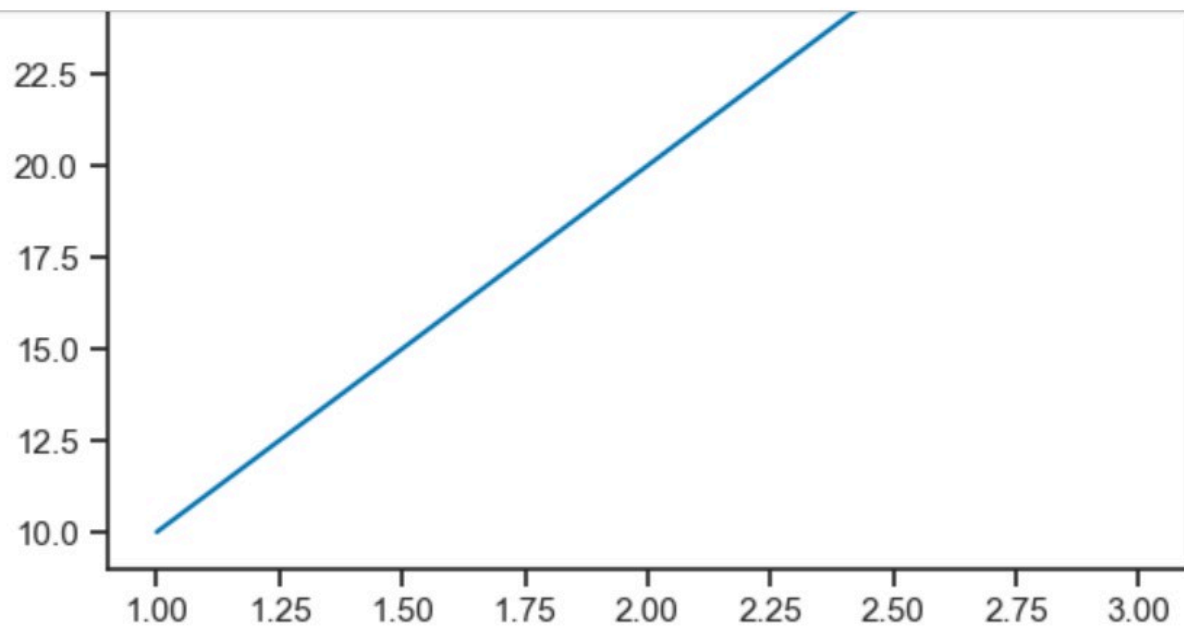
plt.title("Pie")
plt.tight_layout() #Multiplt plot

(plt.show)
plt.savefig("sales_chart.png") #To save the pic
  
```



```
[119]: import matplotlib.pyplot as plt
plt.plot([1,2,3],[10,20,30])
plt.title("Sales Trend")
plt.savefig("sales_chart.png")
plt.show()
```





```
[121]: #Save with high quality
plt.savefig(
    "chart.png",
    dpi=300
)
```

<Figure size 640x480 with 0 Axes>